PROJECT: Agentic AI for Continuous Software Quality Improvement (Replay-Validated)

INPUTS

1. Commits and diffs
2. Issues, bug reports and requirements
3. Pull requests and review comments
4. Test files and docs (read from the code at each commit)
5. CI runs (test duration, build time, pass/fail). Conditional: most public repos do not have this data.

OUTPUTS AND GROUND-TRUTH RULES

(T = the moment of replay. The system sees only data before T. The rule is checked against data after T, within a fixed window, e.g. 6 months.)

1. Detect a risky change (possibly introducing a bug)
   Ground truth: The flagged commit is correct if the algorithm later traces a bug-fix commit back to it (the fix modifies lines the flagged commit introduced).

2. Missing test coverage
   Ground truth: The flag is correct if the change shipped without test changes and a test for that code is added later, or a later bug fix touches that code.

3. Similar past issue detection in new issues (so the previous fix can be reused)
   Ground truth: The match is correct if the new issue is later closed as a duplicate of, or explicitly linked to, one of the top-k past issues the system returned.

4. Similar bug detection directly from diffs (feeds Output 1)
   Ground truth: The match is correct if the diff resembles a past bug-introducing change AND the flagged change is later confirmed buggy by algorithm, or a developer

later references the matched past bug in the fix.

5. Outdated documentation detection

Ground truth: The flag is correct if the doc is updated in a later commit to match the flagged code change, or a documentation issue is opened for it.

6. Test time / build time regression (conditional on Input 5)

Ground truth: The flag is correct if the increase against the baseline (e.g. more than 25% above the median of the previous 20 runs) persists over the next N runs and is later reverted, fixed, or reported as an issue.

7. Requirement vs implementation mismatch

(requirement = issue text, PR description or commit message)

Ground truth: The flag is correct if the linked issue is later reopened, a follow-up issue or PR says the fix was incomplete ("still happens", "also needs..."), or the change is reverted. Ground truth is noisy, so also manually review a sample of about 30 cases.

TIERS

Core: Inputs 1-4; Outputs 1, 2, 3, 4

Secondary: Outputs 5 and 7

Stretch: Input 5 and Output 6 (needs a repo with CI history)

EVALUATION (applies to all outputs)

- Metrics: precision, recall, F1, lead time (how early the warning came)
- Baselines: random, and "most-changed files" heuristic
- Leakage rule: no data after T may reach the system (enforced in code)
- Memorization check: include repos or periods after the model's training cutoff